

CEK8903_PIQ_DDR4_ONBOARD

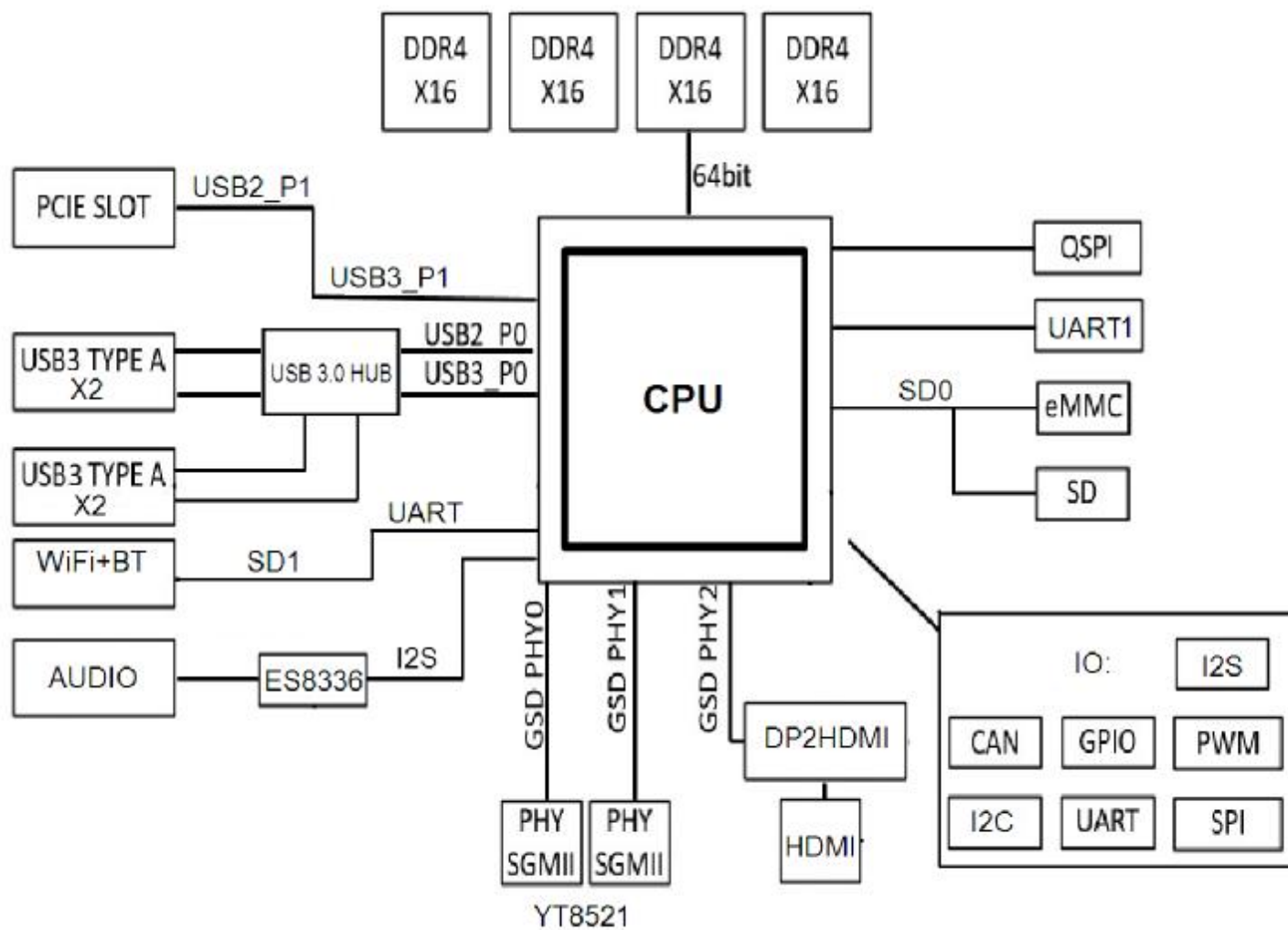
PHYTIUM
CONFIDENTIAL

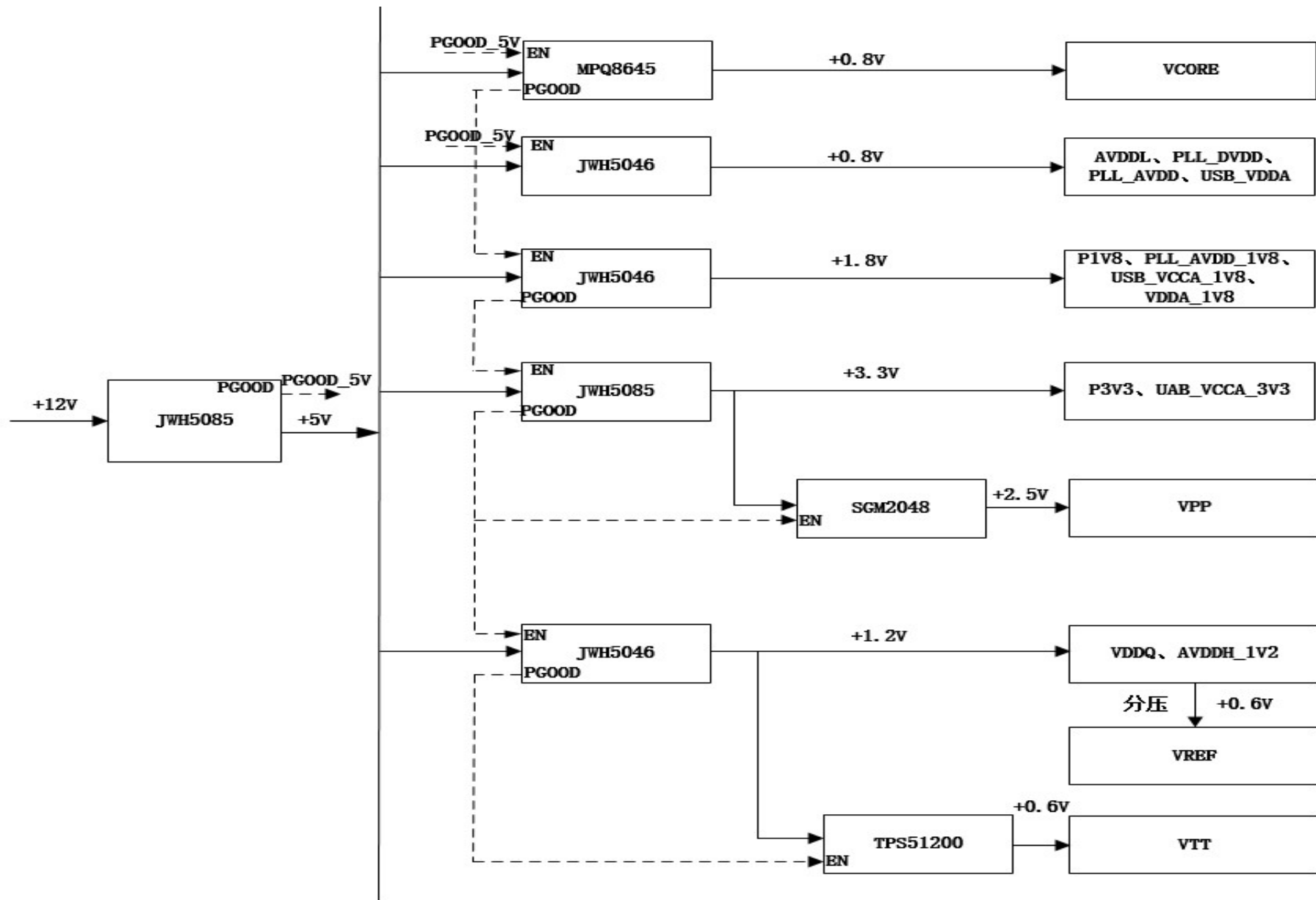
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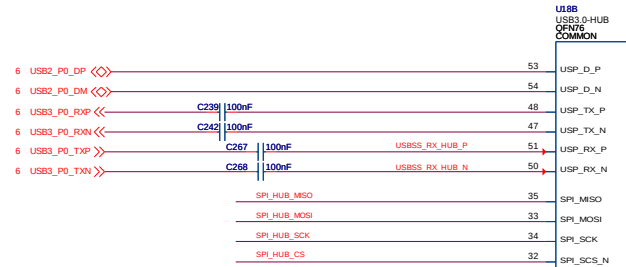
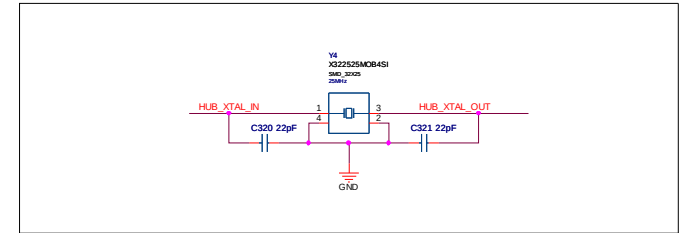
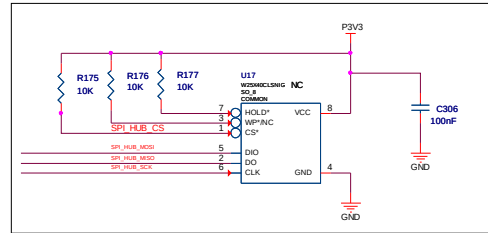
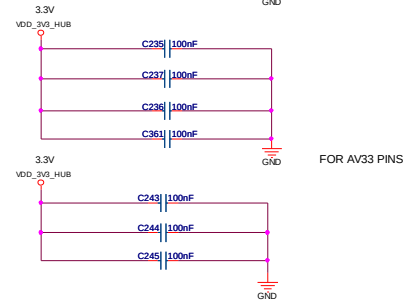
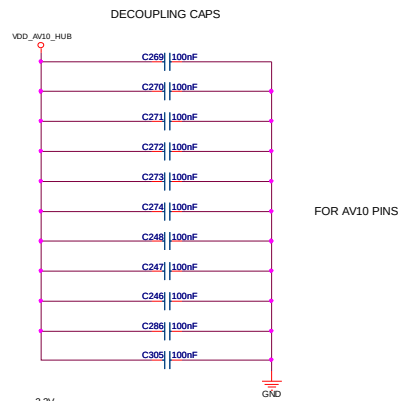
Page	Index
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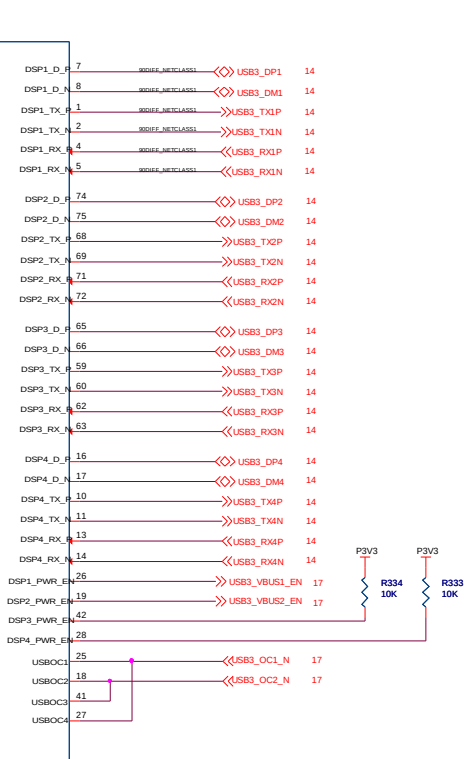
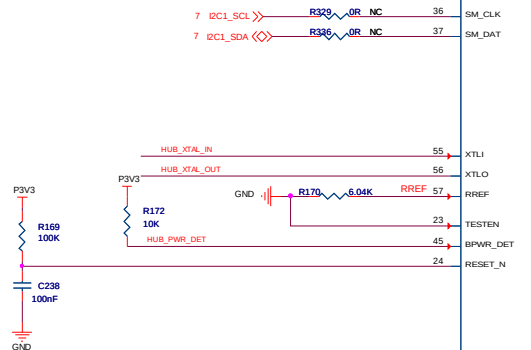


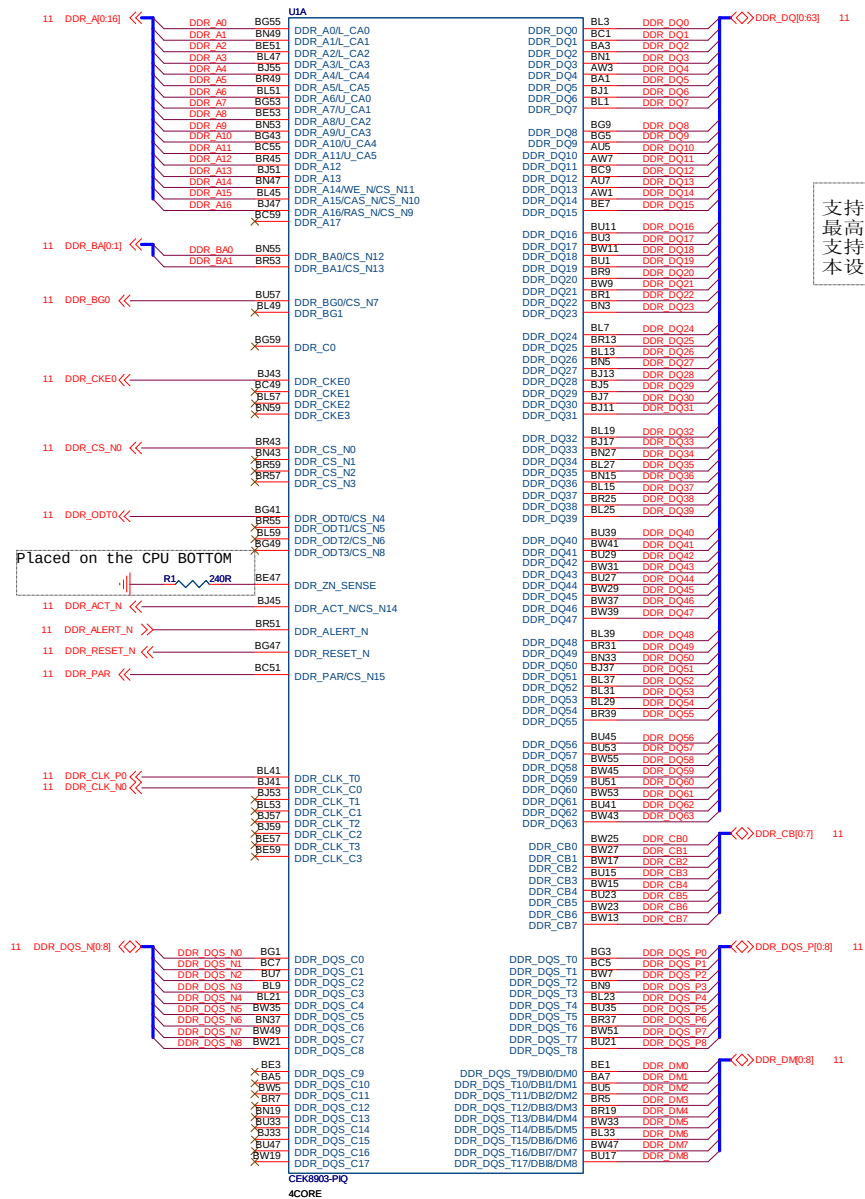
本页电路为新增，目的是将4个USB口全升级为USB3.0

USB3.0 HUB x4 PORTS



I2C1线路与HUB之间增加0欧电阻





支持器件: DDR4: x4、x8、x16; LPDDR4: x16、x32,
最高支持72个数据位。
支持全位宽模式和半位宽模式 (仅支持低32位数据和低4位ECC)
本设计采用DDR4 X16颗粒

CECport Firefly Workshop

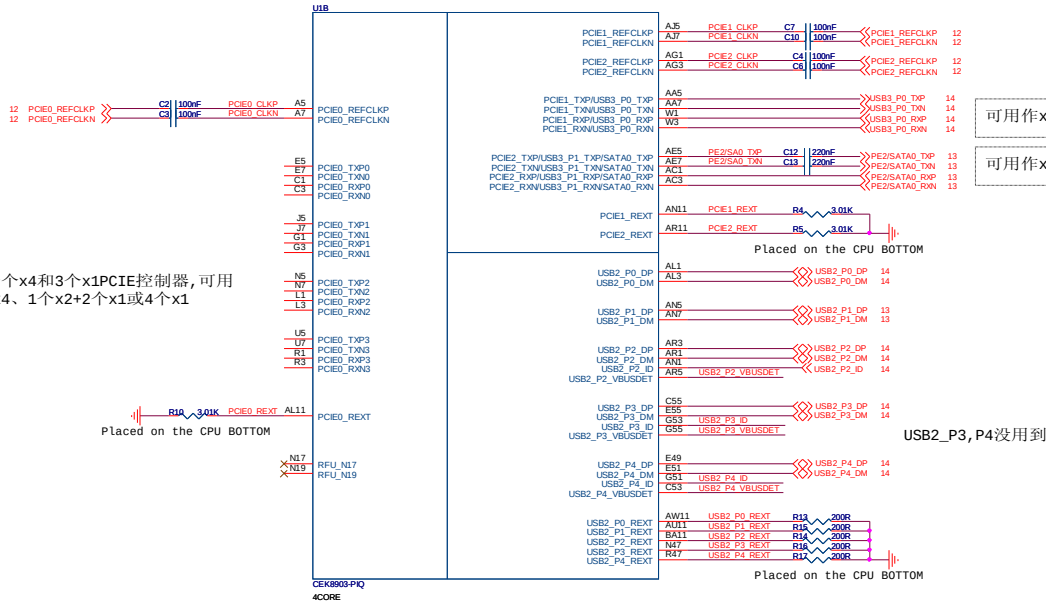
CEK8903-PIQ

TitleCPU_MCU

SizeCustomDocument NumberSCH20221128RevV3.1

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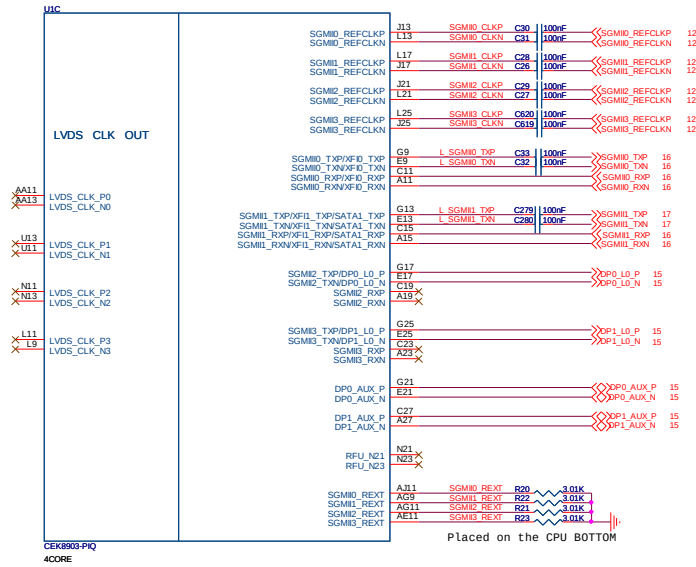
包含一个x4和3个x1PCIE控制器,可用
作1个x4、1个x2+2个x1或4个x1

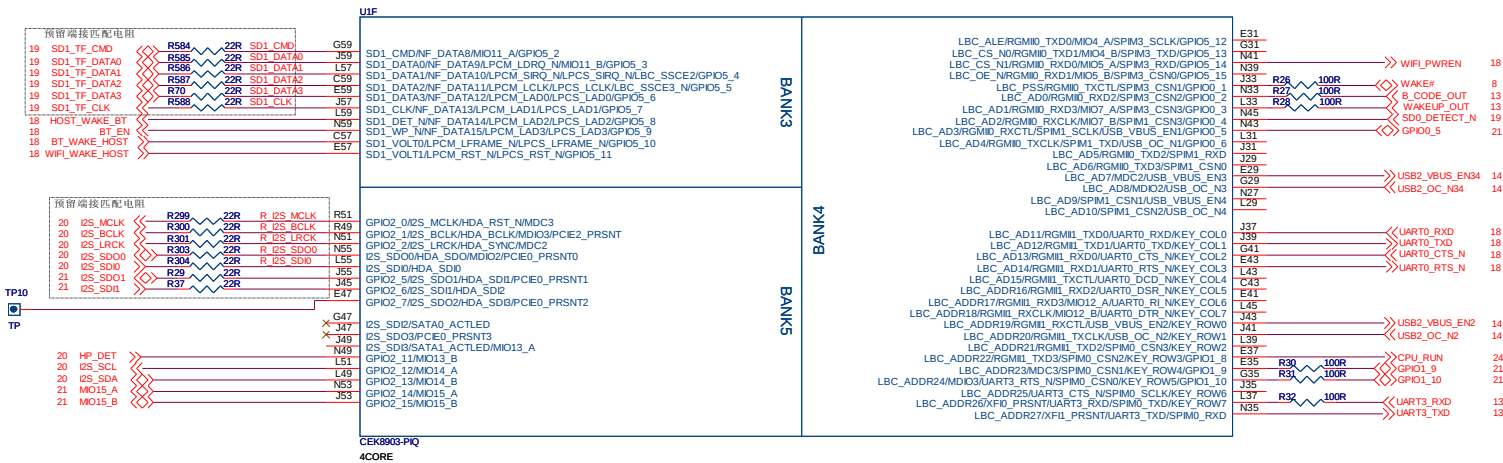


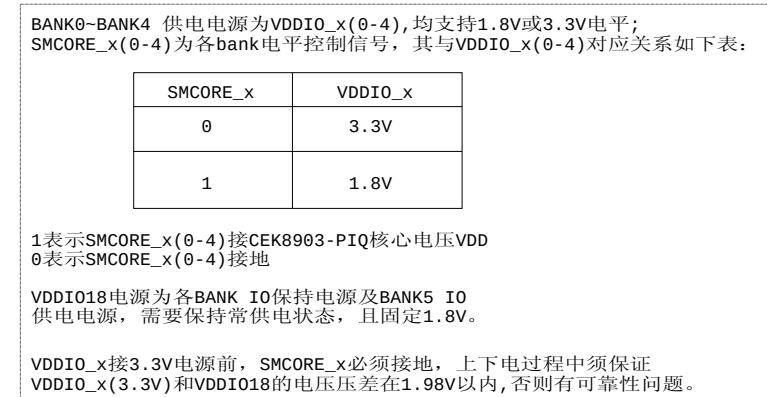
CEK8903-PIQ包含了4个MAC控制器MAC0/1支持XFI和SGMII;
MAC2/3支持SGMII和RGMII,且共用MDIO。

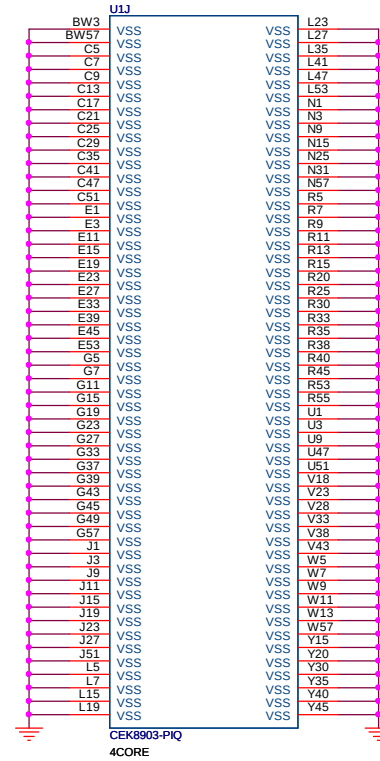
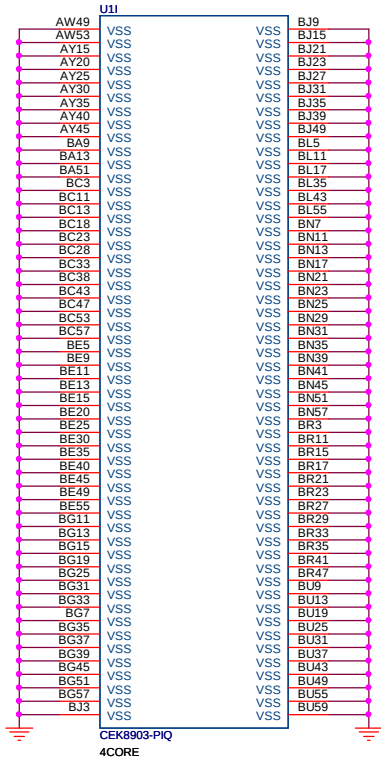
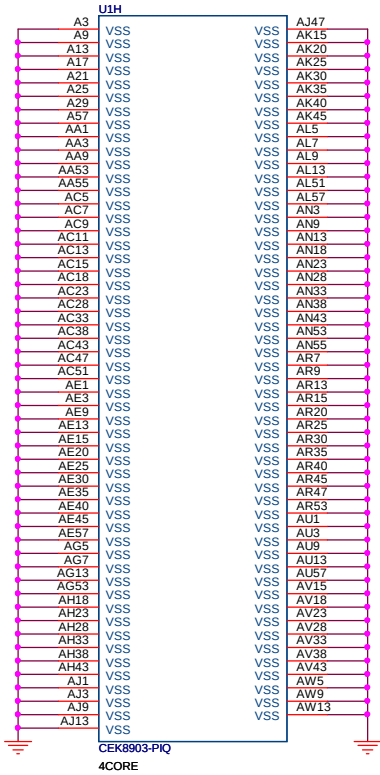
MDIO管理接口与MAC端口对应关系

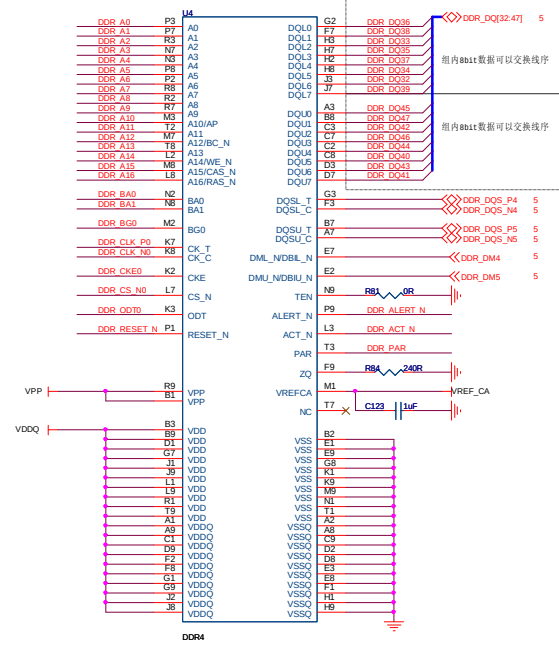
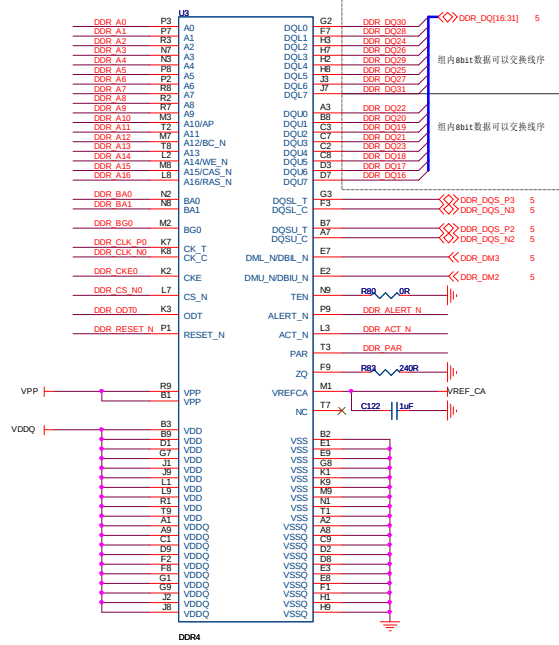
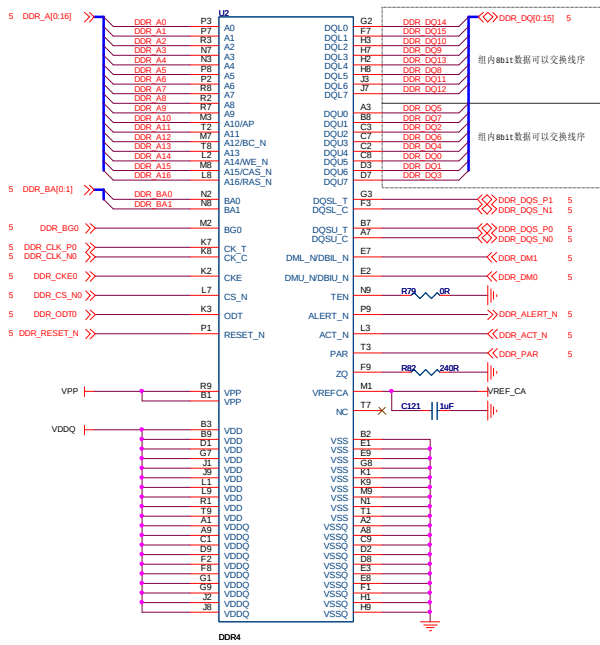
MDIO管理接口	MAC端口类型
MDIO0/MDC0	SGMII0
MDIO1/MDC1	SGMII1
MDIO2/MDC2	SGMII2/RGMII0
MDIO3/MDC3	SGMII3/RGMII1



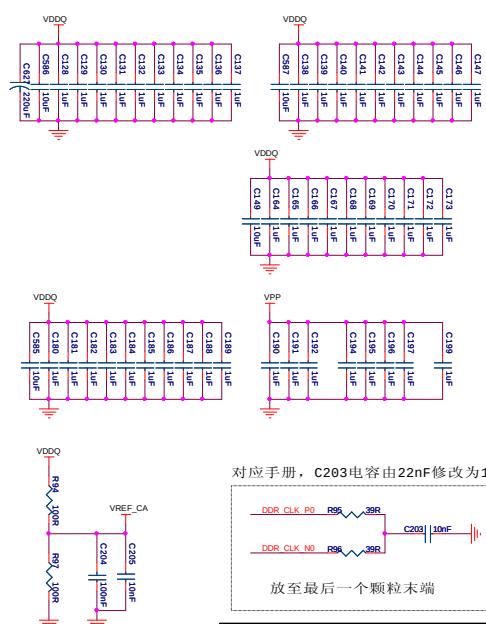
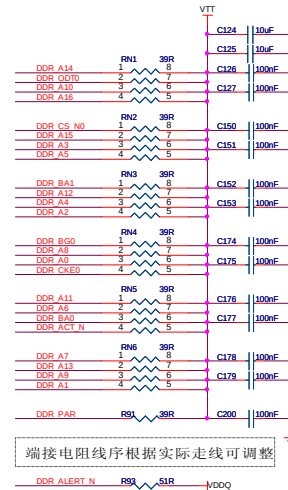
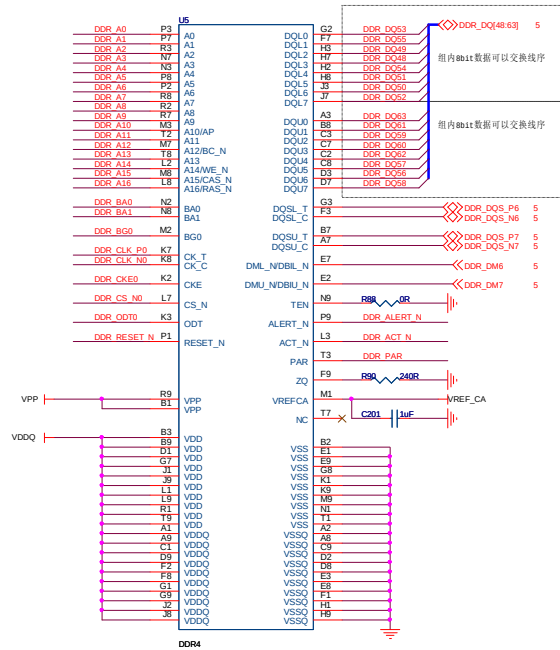




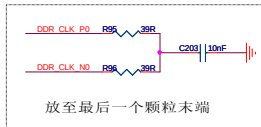




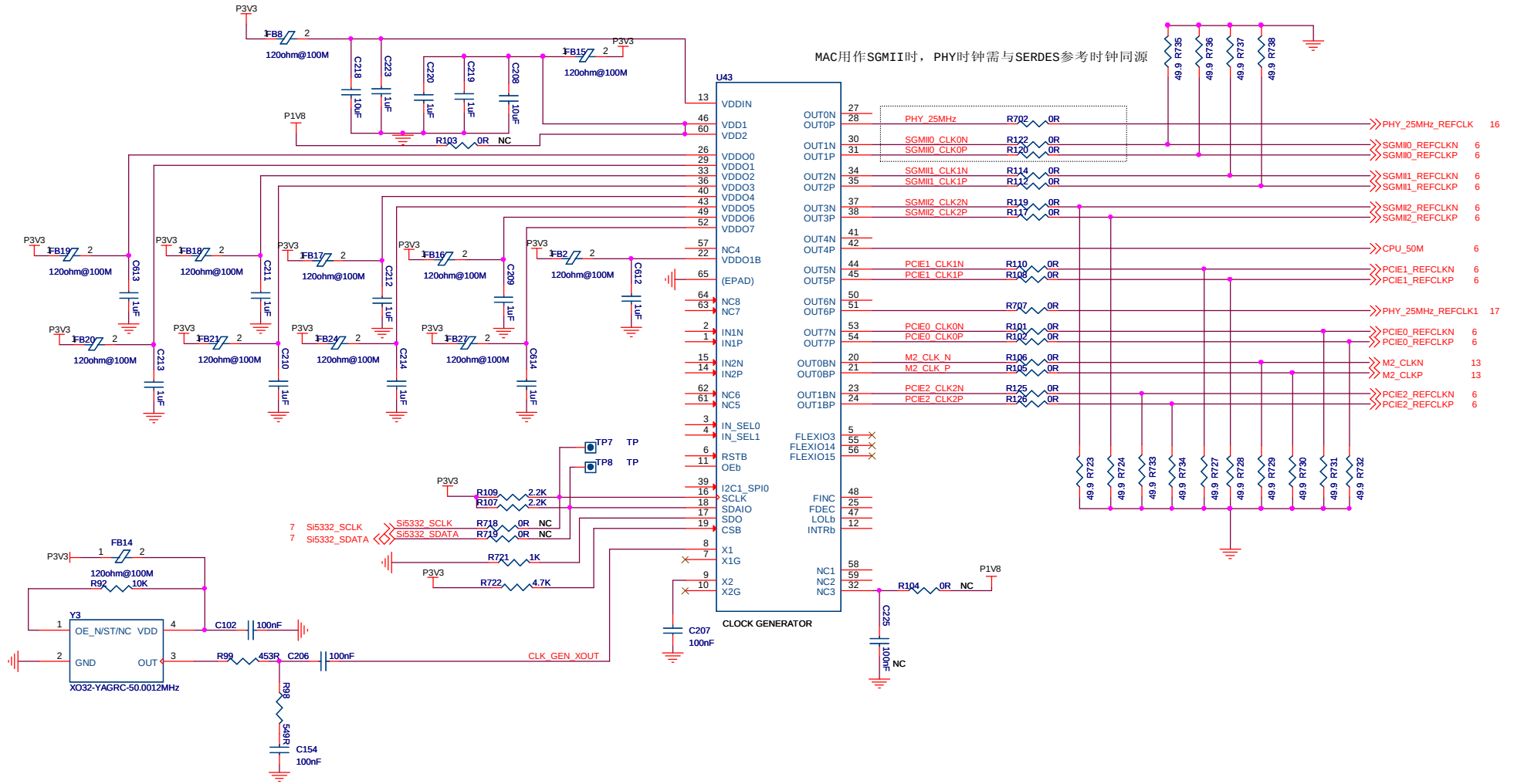
除ECC外，每个slice之间可以交换



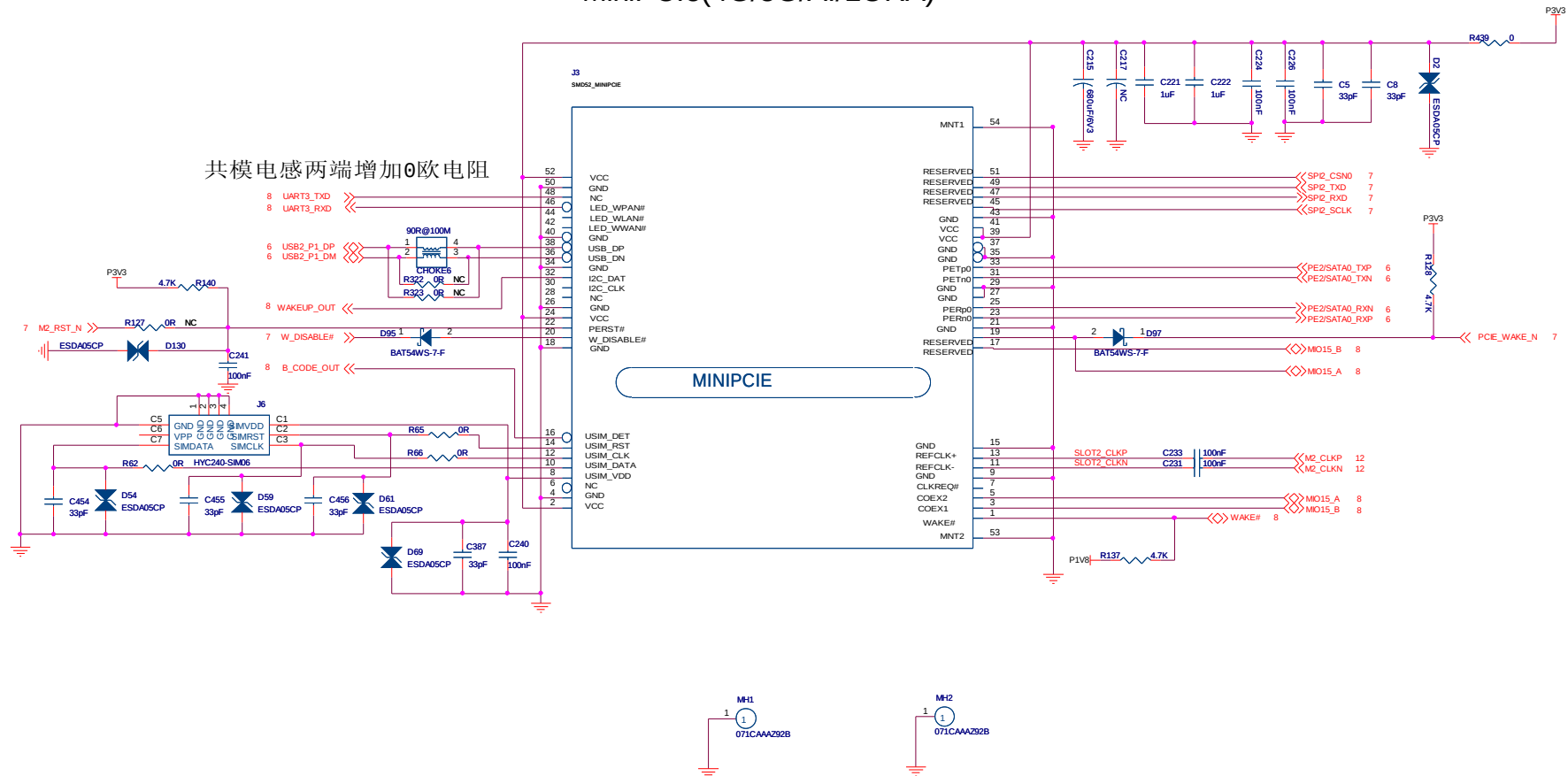
对应手册，C203电容由22nF修改为10nF



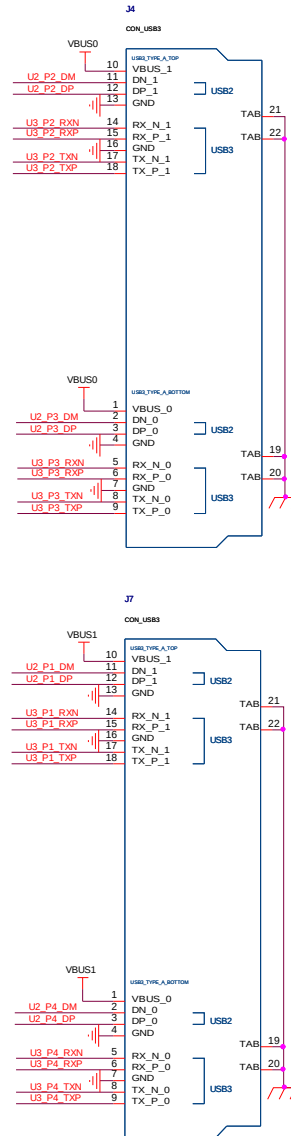
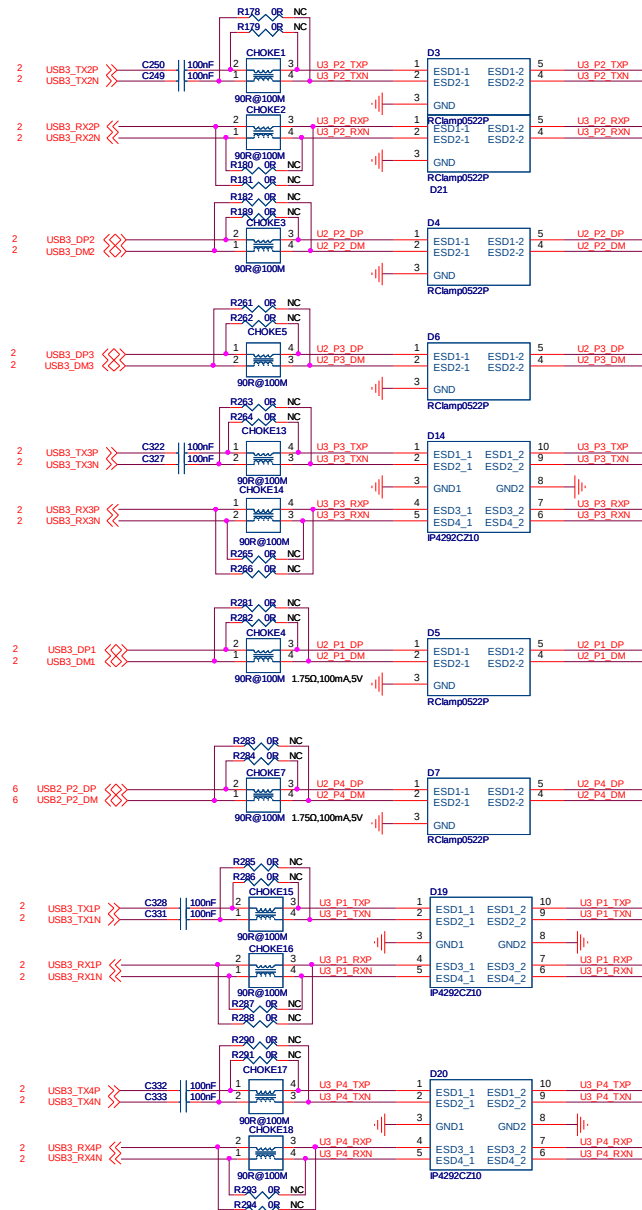
放至最后一个颗粒末端



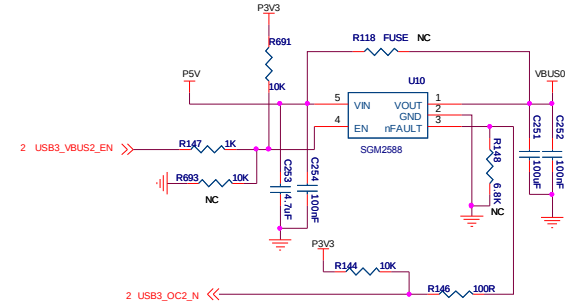
MiniPCIe(4G/5G/AI/LORA)



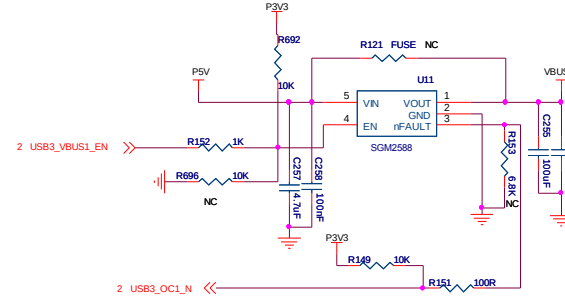
共模电感两端增加0欧电阻



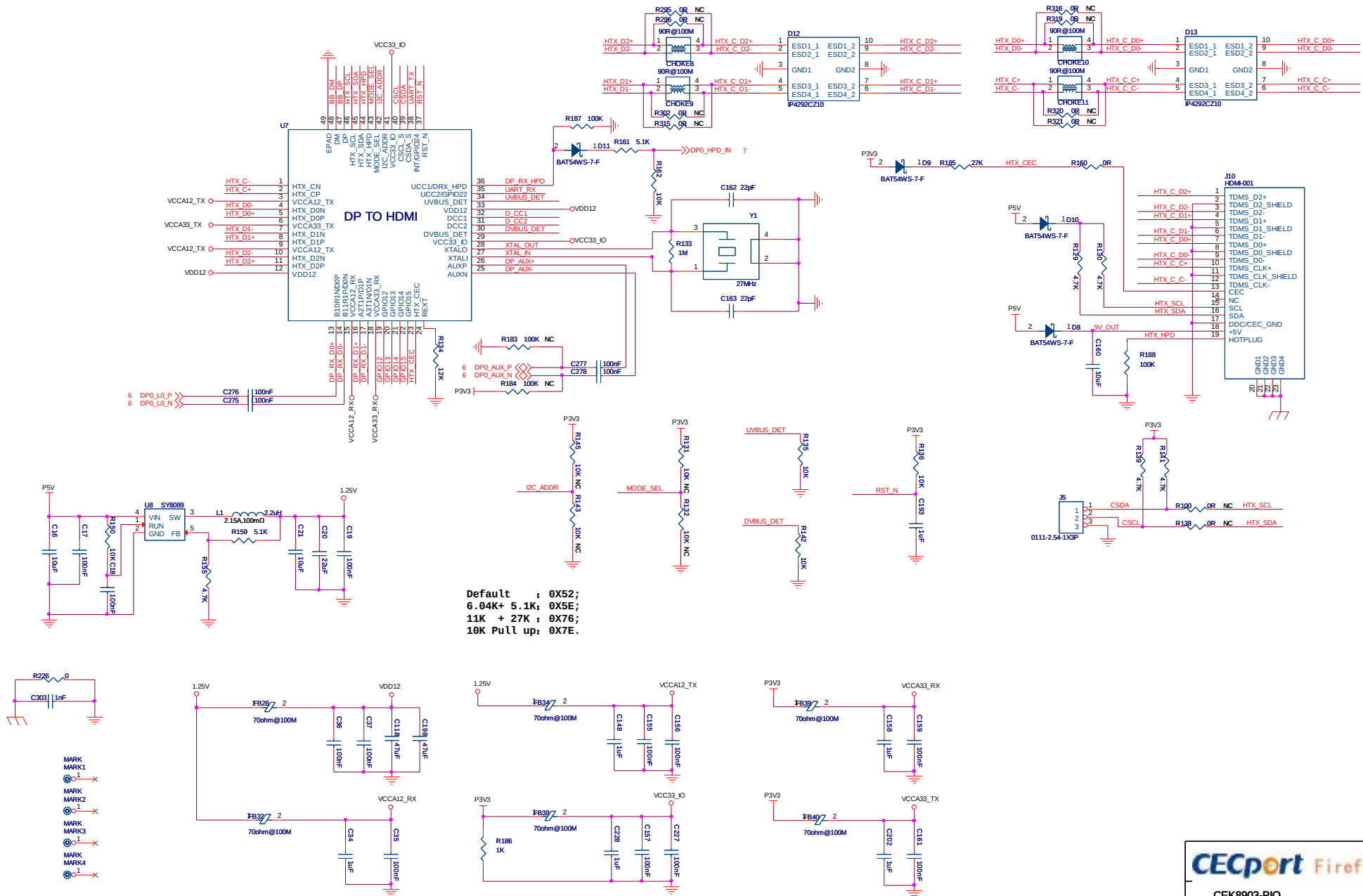
USB3.0和USB2.0做HOST时，VBUS_EN和OC_N可用GPIO代替
用户如需上电直接使能USB，则上拉使能即可（本设计预留）

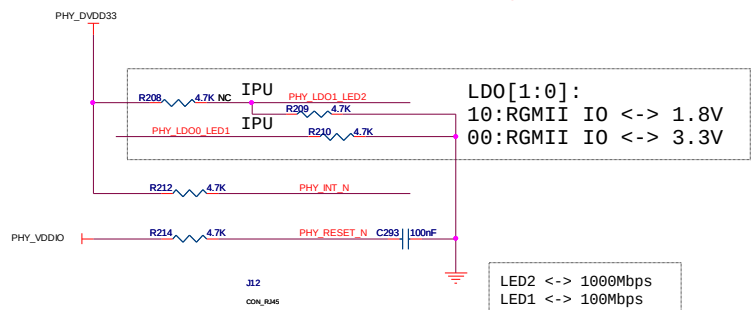
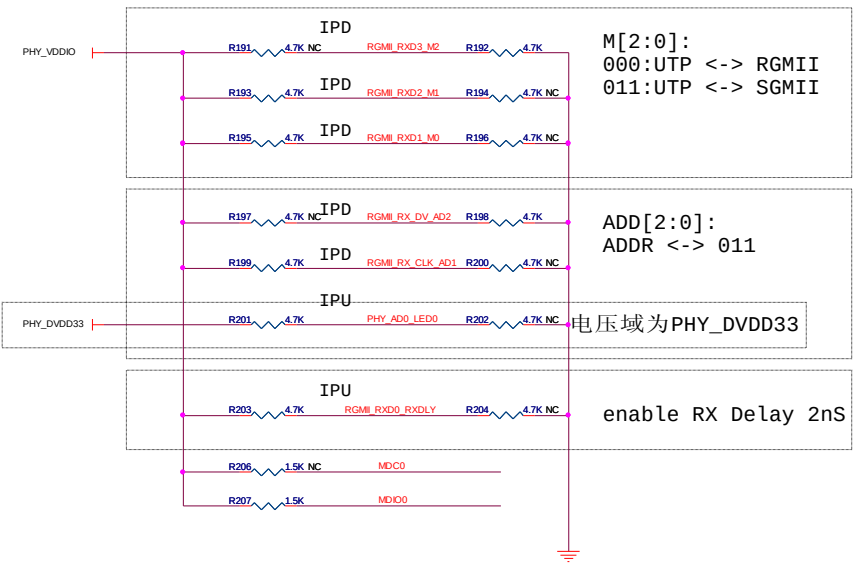
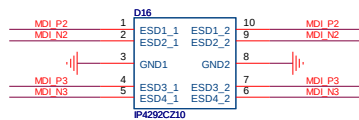
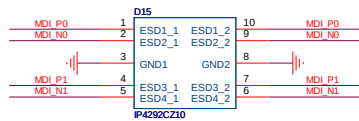
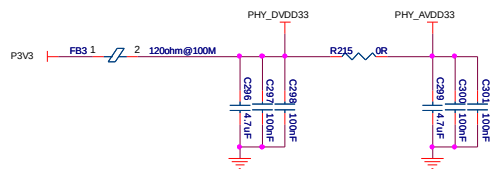


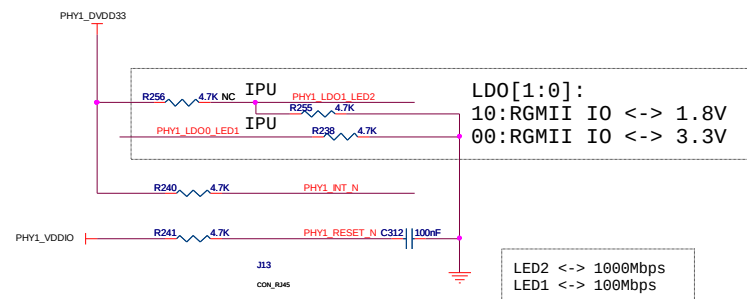
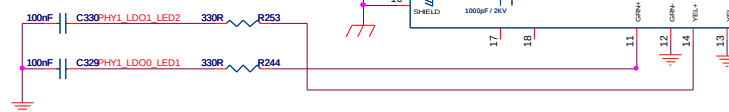
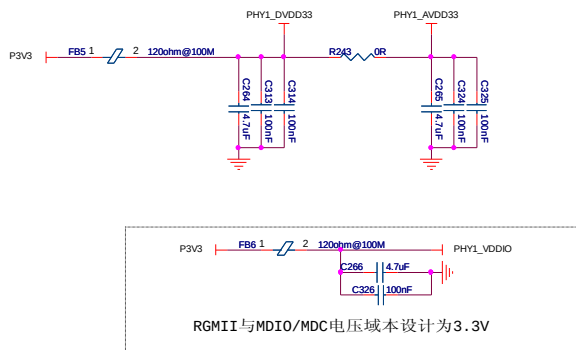
用户如需上电直接使能USB，则上拉使能即可（本设计预留）

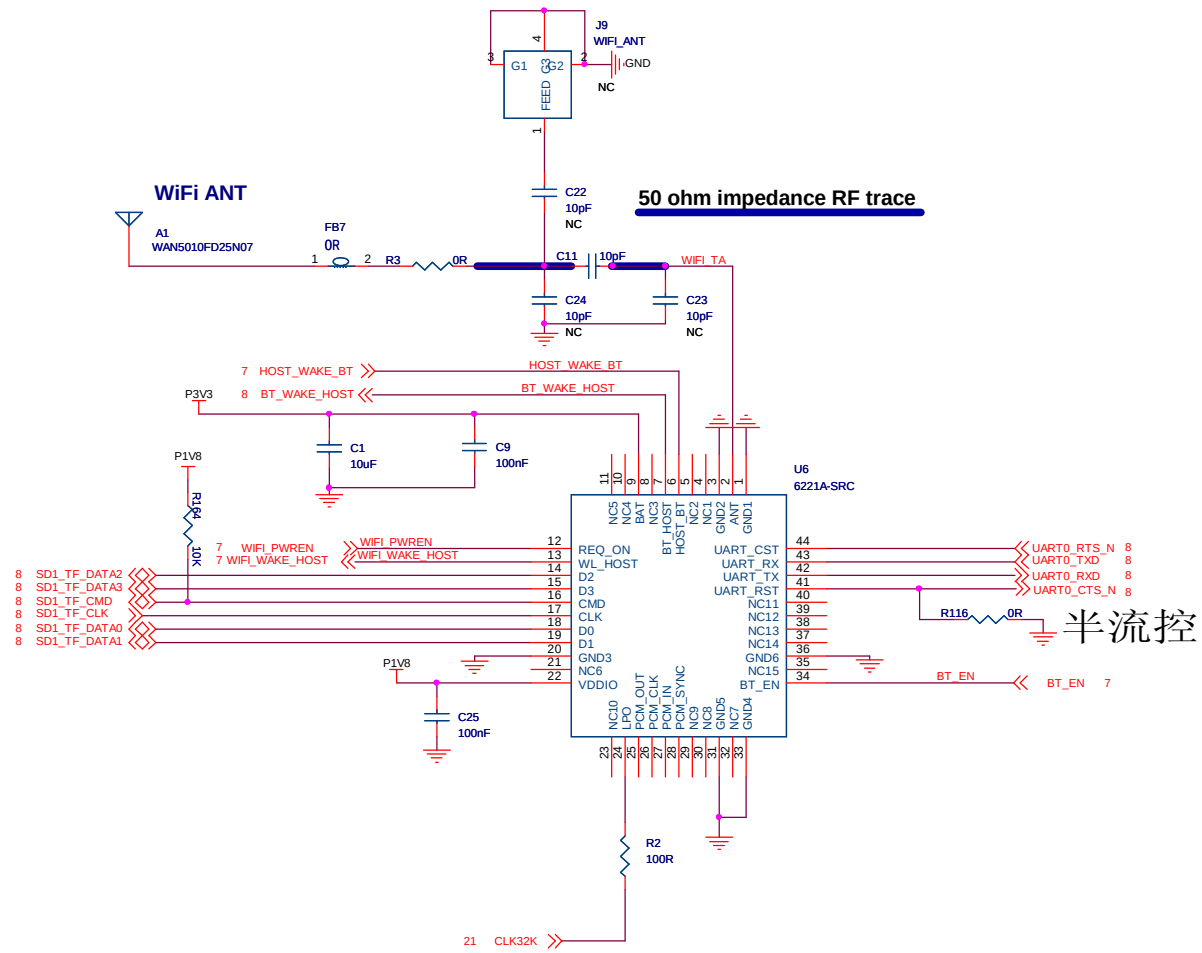


共模电感两端增加0欧电阻

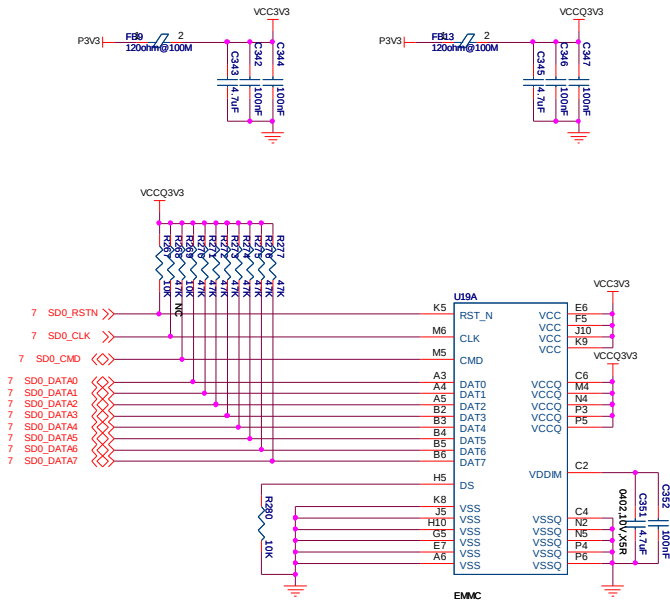
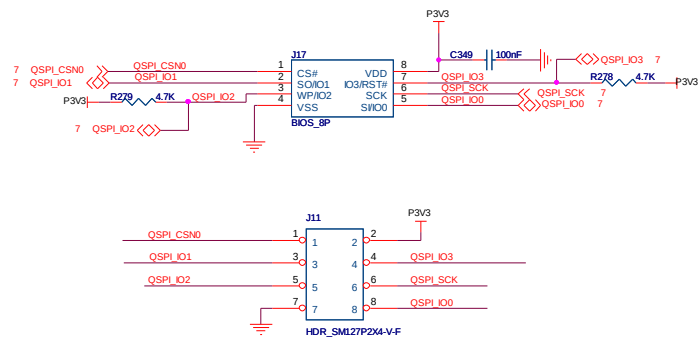






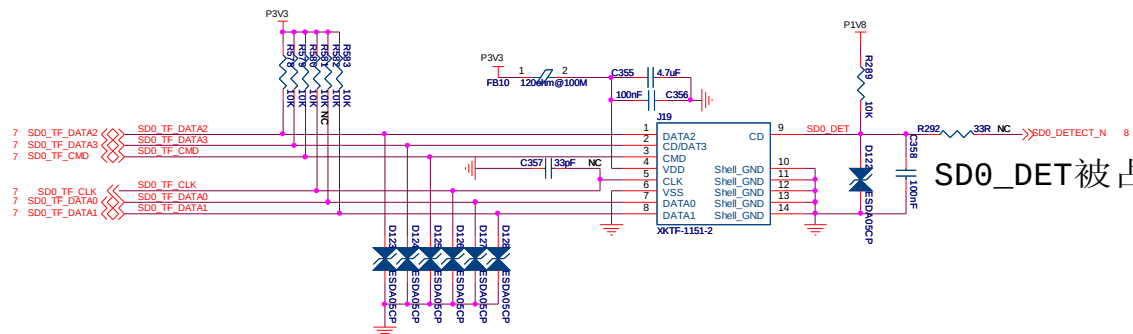


QSPI BIOS



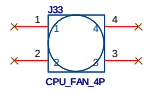
U19B		
A1	NC1	G1
A2	NC2	G2
A8	NC3	G3
A9	NC5	G12
A10	NC4	G13
A11	NC5	G14
A12	NC6	H1
A13	NC7	H2
A14	NC8	H3
B1	NC9	H12
B7	NC10	H13
B8	NC11	H14
B9	NC12	J1
B10	NC13	J2
B11	NC14	J3
B12	NC15	J12
B13	NC16	J13
B14	NC17	J14
C1	NC18	K1
C3	NC19	K2
C5	NC20	K3
C7	NC21	K12
C8	NC22	K13
C9	NC23	K14
C10	NC24	L1
C11	NC25	L2
C12	NC26	L3
C13	NC27	L12
C14	NC28	L13
D1	NC29	L14
D2	NC30	M1
D3	NC31	M2
D4	NC32	M3
D12	NC33	M7
D13	NC34	M8
D14	NC35	M9
E1	NC36	M10
E2	NC37	M11
E3	NC38	M12
E12	NC39	M13
E13	NC40	M14
E14	NC41	N1
F1	NC42	N3
F2	NC43	N6
F3	NC44	N7
F12	NC45	N8
F13	NC46	N9
F14	NC47	N10
K7	NC48	N11
K8	RFU1	N12
E5	RFU2	N13
A7	RFU3	N14
E8	RFU4	P1
E9	NC102	P2
F10	VSF1	P7
G10	VSF2	P8
G11	VSF3	P9
G12	VSF4	P11
G13	VSF5	P12
G14	VSF6	P13
H1	VSF7	P14

EMMC



SD0_DET被占用，使用GPIO0_4模拟

ESD靠近TF卡槽放置



5V FAN

7 TACH2

7 PWM1

PSV3V3

R123 10K

R154 10K

R156 10K

R157 10K

R158 10K

R167 10K

R168 1K

R171 1K

Q3 2N7002

Q4 2N7002

D23 CMH25242B

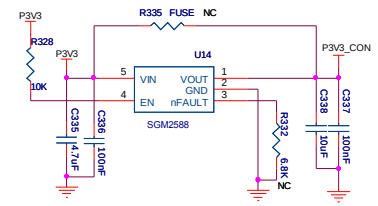
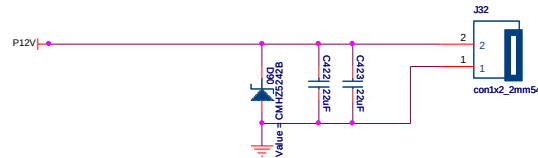
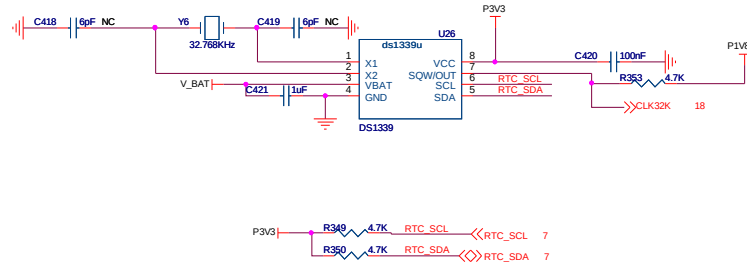
PSV

PSV

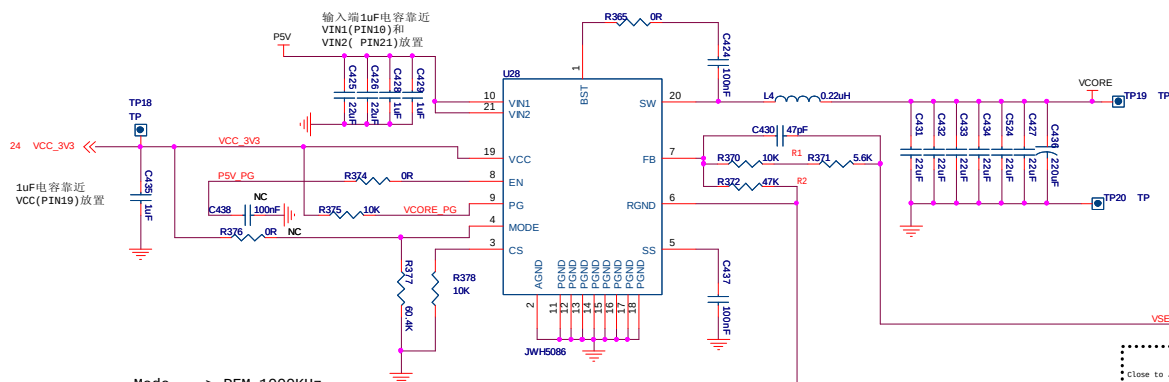
PSV

PSV

CON4 smd 1.25mm

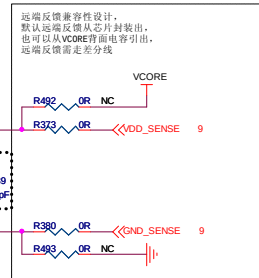


24,25 PSV_PG TP
24 VCORE_PG TP

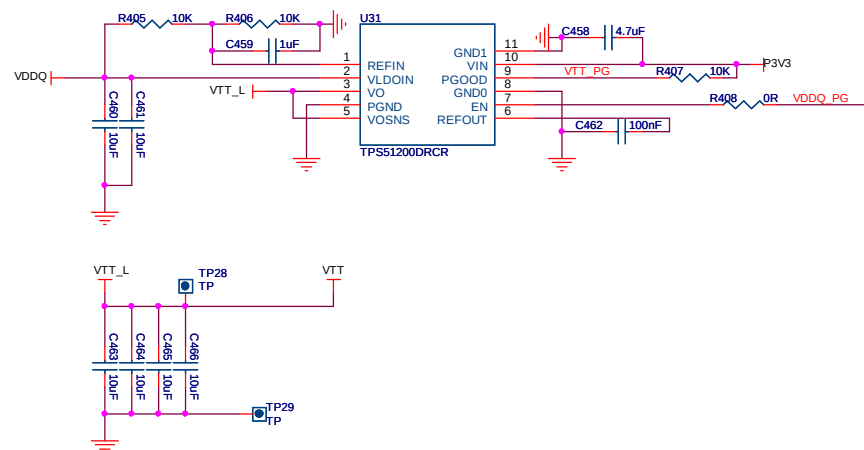
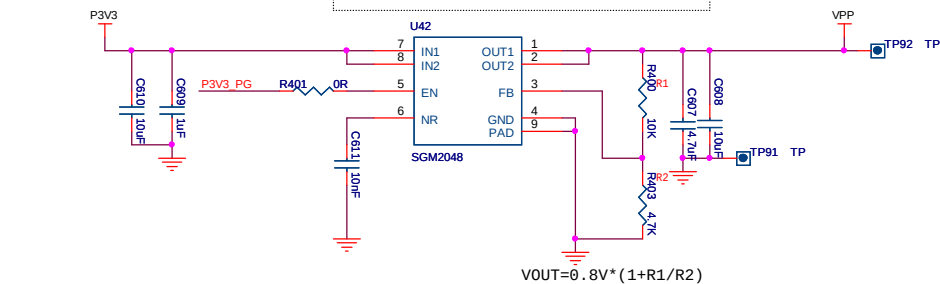
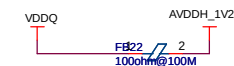
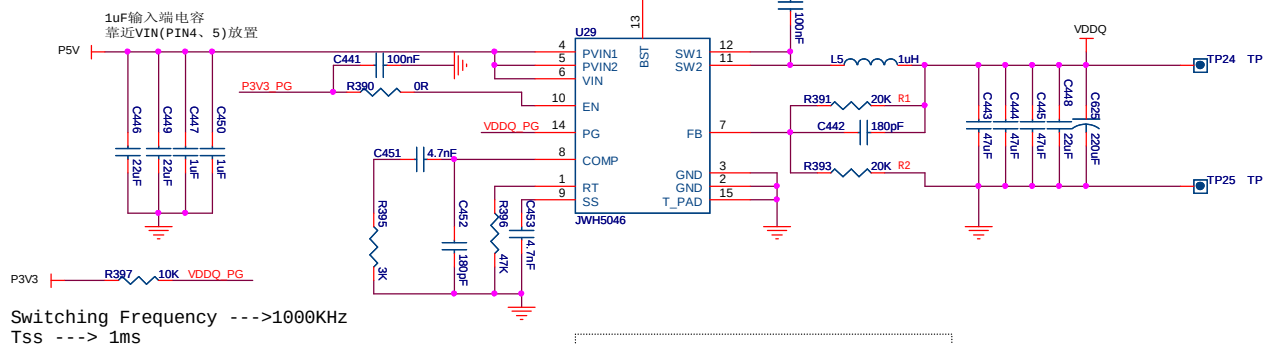


24 VCC_3V3
1uF电容靠近
VCC(PIN19)放置

Mode ---> PFM 1000KHz
Ilim ---> 12A
Tss ---> 1.14ms



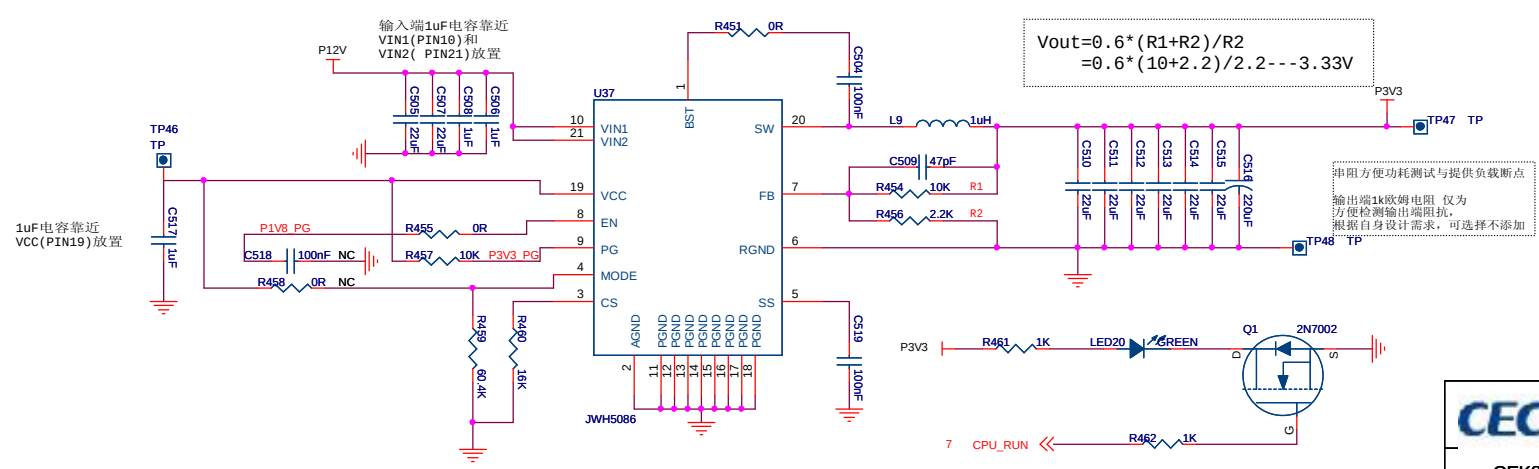
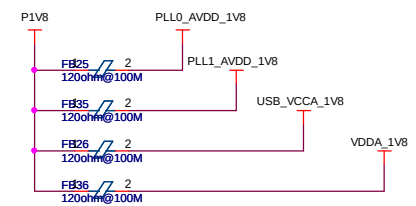
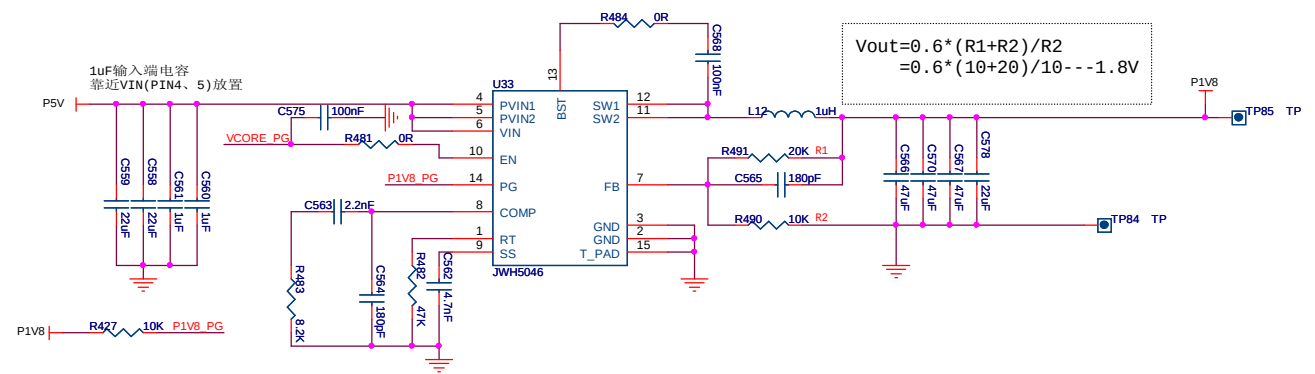
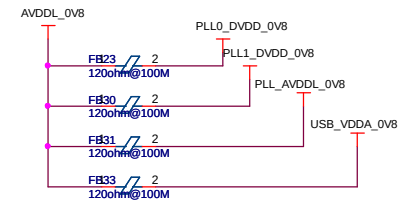
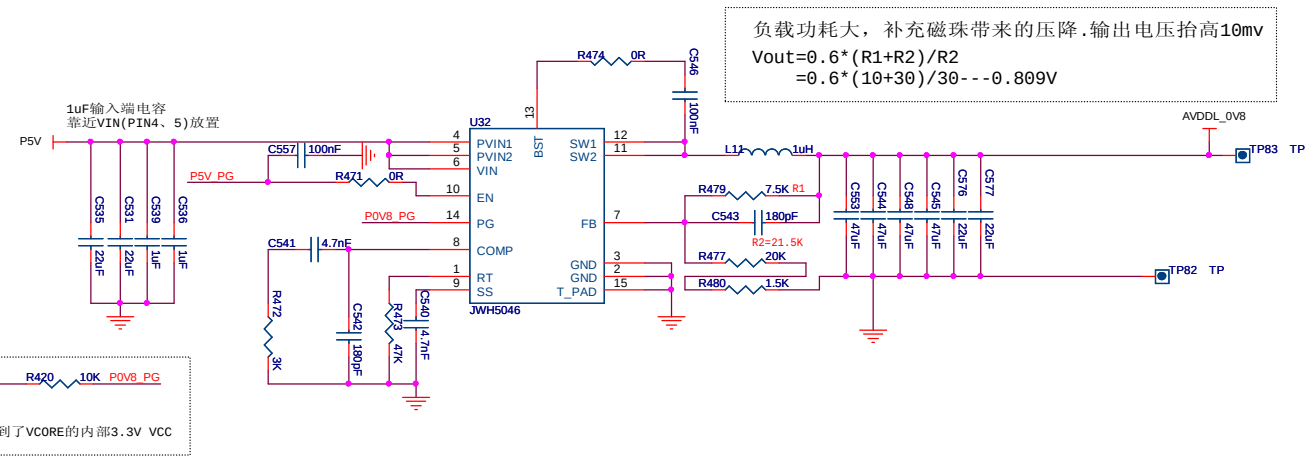
24 P3V3_PG >> TP21 TP



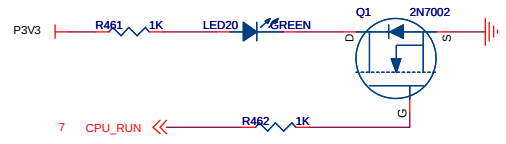
TP31 TP
TP32 TP
TP33 TP

22 VCORE_PG
22.25 P5V_PG

23 P3V3_PG
22 VCC_3V3



Mode ----> PFM 1000KHz
 Ilim ----> 9.5A
 Tss ----> 1.14ms



CECport Firefly Workshop

CEK8903-PIQ

Title

PWR_P0V8&P1V8&P3V3

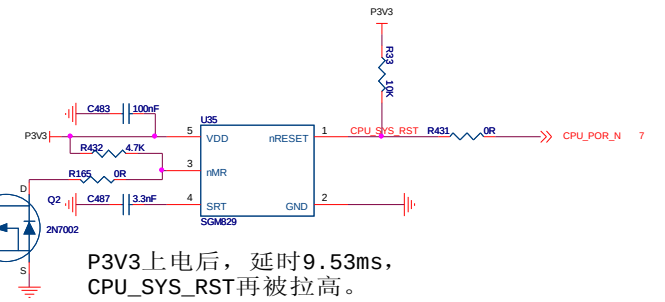
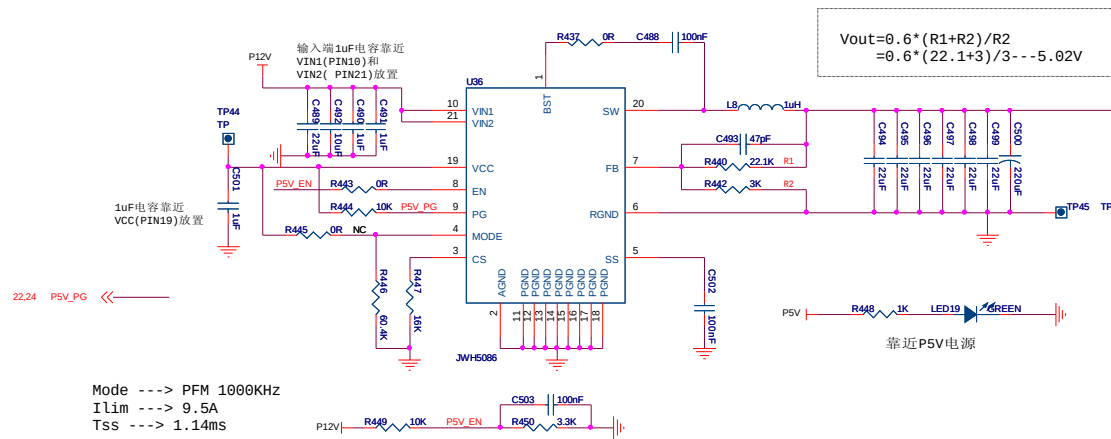
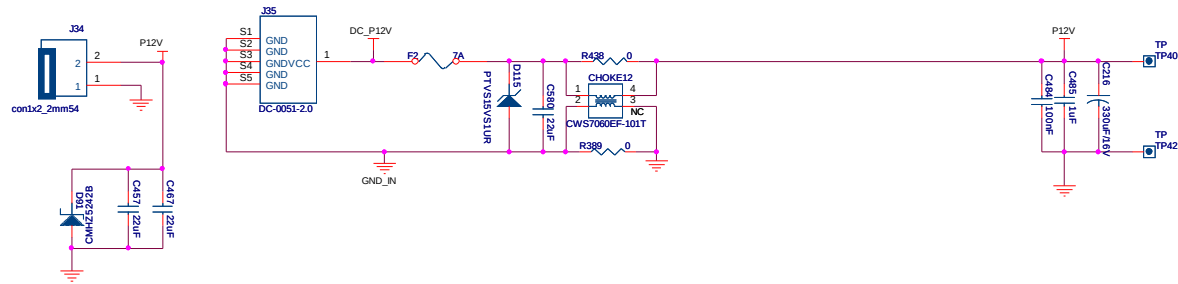
Size A3

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$$T_D(\mu s) = (2.8 \times 10^6) \times C_3(\mu F) + 290 \mu s$$

本参考设计有复位按钮、上电复位、FPGA复位三种方式接入了CPU的POR_RST：

1. 客户如需使用FPGA做LBC锁存或其它用途，建议使用FPGA控制上电复位，本设计有预留；
2. 无FPGA场景，用户可以选择电源PG信号控制上电时序，使用复位芯片上电复位；
3. 本设计与参考板对应，为满足电源PG信号控制上电时序示例的同时，验证LBC启动功能，使用FPGA复位CPU（FPGA锁存需要优先CPU复位）。

